A Continuous Moment of Fluid Algorithm for simulating the interaction of multiphase and multimaterial fluid(s) with flexible solid(s) *

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ABSTRACT: A new Eulerian method is presented for simulating complex three dimensional and three dimensional axisymmetric flows consisting of any number of materials, to include compressible fluids, incompressible fluids, ice, melt, vapor, and solids undergoing elastic/plastic deformation. The interfaces between materials are discretized using the continuous moment of fluid method.

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1 Introduction

The ability to simulate multiphase and multimaterial high and low speed flows, to include the interaction of fluid(s) and elastic material(s), has a number of applications: bubble growth and collapse interacting with flexible surfaces[61, 62, 56], wetting dynamics of liquid droplets on deformable membranes[39, 29, 20, 28], passive aircraft anti-icing[36], design of breathable masks[17], underwater explosions near structures[34, 64, 12, 21, 24, 57, 45], hydroelastic applications[38], vibration of pipes due to internal boiling flow[5], ultrasonic leak detection in nuclear power plants[14], and the measurement of the tensile strength properties of colliding and deforming structures[15].

We are not aware of previously reported research on the robust numerical simulation of the interaction of multiphase flows with elastic materials, to include incompressible materials, weakly compressible materials, and/or strongly compressible materials. The research that we report here is a continuation of research that started with the following article by Jemison et al[26]. In Jemison et al's research, a novel adaptive, parallel, asymptotically preserving[31, 9, 10, 46], moment-of-fluid method was presented for simulating two phase flows to include incompressible, low speed, and high speed flows. The method in Jemison et al was a collocated generalization of the projection method for incompressible flows[25] in order to simulate both compressible (with shock waves) and incompressible flows.

In the present work, we have improved upon [26] in the following important ways:

1. the present method is now a staggered grid (and conservative) method instead of colocated. See Figures 1, 2, and 3. The following research for incompressible flows underscore the importance of a staggered grid[33, 50, 48, 23] for projection methods. We also report recent work by Park and Son[42] in which they developed a (non-conservative) staggered grid method for compressible flows in which the acoustic time step constraint was avoided.

2. the present method represents complex deforming interfaces using the Continuous Moment-of-Fluid method[63] (CMOF), instead of the Moment-of-Fluid (MOF) method[27, 2, 3]. The CMOF method[63] removes a "checkerboard instability" that appears for surface tension driven flows.

3. We have developed a conservative, staggered grid, algorithm for solving the equations associated with standard elastic materials and materials undergoing plastic deformation. Our new algorithm is an improvement on the algorithms reported in [50, 48, 42] so that we can simulate both low and high speed fluid structure interaction problems. Our new method can simulate incompressible elastic materials or compressible elastic materials with shock waves.

In Table 1, we compare and contrast our new method with the current state of the art.

Author(s)	incompressible fluids	compressible fluids	multiphase flow	multimaterial flow	shock waves	Coupling Method	Interface Representation	Complex topology
Wardlaw and Luton [57]	No	Yes	No	Yes	Yes	partitioned	ALE (fluid interfaces) Lagrangian (structure)	Yes
Tran and Udaykumar [51]	No	Yes	No	Yes	Yes	Explicit	Particle Level Set	Yes
Robinson-Mosher et al. [44]	Yes	No	No	No	No	Monolithic	Connected Markers	No
Gretarsson et al. [22]	No	Yes	No	No	Yes	Monolithic	one dimension	No
Sugiyama et al. [50]	Yes	No	No	No	No	pressure and velocity	Volume of fluid	yes
Banks et al. [7]	Yes	No	No	No	No	Partitioned	Body fitted	No
Patkar et al. [43]	No	Yes	No	No	Yes	Monolithic	Connected Markers	No
Banks et al. [8]	Yes	No	No	No	No	Partitioned	Body fitted	No
Shin et al. [48]	Yes	No	Yes	Yes	No	pressure and velocity	level contour reconstruction	Yes
Mokbel et al. [39]	Yes	No	No	Yes	No	Monolithic	Hybrid Diffuse and ALE	No
Kloppe and Aland [30]	Yes	No	No	Yes	No	Monolithic	Hybrid Diffuse and ALE	No
Gao et al. [19]	No	Yes	No	Yes	Yes	explicit moving mesh ALE	moving mesh ALE	Yes
Ferrari et al. [18]	No	Yes	Yes	Yes	Yes	explicit non-partitioned	Diffuse interface	Yes
Wilfong et al. [59]	No	Yes	Yes	Yes	Yes	explicit non-partitioned	Diffuse interface	Yes
Park and Son [42]	Yes	Yes	Yes	Yes	No	pressure and velocity	Coupled level set volume of fluid	Yes
Present Article, Estebe et al	Yes	Yes	Yes	Yes	Yes	pressure and velocity	Continuous Moment of Fluid	Yes

Table 1: Recent numerical methods with fluid interaction with deformable structures. Legend: ALE (Arbitrary Lagrangian Eulerian), "pressure and velocity coupling" refers to the asymptotically preserving methods which reduce to the projection method in the zero Mach number limit[42, 50, 48], "Partitioned coupling" refers to alternating solving the fluid equations with elastic material boundary conditions and then solving the elastic material equations with fluid boundary conditions.

1.1 Motivation for enhancing our asymptotically preserving algorithm from collocated grid to staggered grid

In this section, we just focus on single material inviscid compressible gas dynamics with a general equation of state and periodic boundary conditions. The derivations here are applicable to the general multimaterial case. The compressible gas dynamic equations[4, 10, 46, 13] are:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{1}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I}) = 0 \tag{2}$$

$$(\rho E)_t + \nabla \cdot (\rho \vec{u} E + p\vec{u}) = 0 \tag{3}$$

$$E = e + \frac{1}{2}\vec{u} \cdot \vec{u} \tag{4}$$

$$p = p(\rho, e) \qquad c^2 = p_\rho + \frac{p_e p}{\rho^2} \qquad e = c_v T \tag{5}$$

Following the "asymptotically preserving" analysis in these articles, [4, 10, 46, 13], it can be shown that our directionally split Cell Integrated Semi-Lagrangian (CISL) algorithm given by Algorithm 1 approaches a discretization consistent with the incompressible flow equations in the zero Mach number limit.

Algorithm 1 Asymptotically Preserving time discretization for gas dynamics

$k \leftarrow 0$
$\vec{u}^{n+1,(k)} \leftarrow \vec{u}^n$
while $k \leq 1$ **do**
  $\rho^{n+1,(k+1)} \leftarrow \rho^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} \vec{u}^{*,(k+1)} \leftarrow (\rho \vec{u})^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u} \otimes \vec{u}^{n+1,(k)}) dt$
  $\rho^{n+1,(k+1)} E^{*,(k+1)} \leftarrow (\rho E)^n + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho E \vec{u}^{n+1,(k)}) dt$
  $p^{*,(k+1)} \leftarrow p(\rho^{n+1,(k+1)}, e^{*,(k+1)})$
  $c^{*,(k+1)} \leftarrow c(\rho^{n+1,(k+1)}, e^{*,(k+1)})$
  $$\begin{pmatrix} \vec{u}^{n+1,(k+1)} \\ p^{n+1,(k+1)} \end{pmatrix} \leftarrow \begin{pmatrix} \vec{u}^{*,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\rho^{n+1,(k+1)}} \\ p^{*,(k+1)} - \Delta t \rho^{n+1,(k+1)} (c^{*,(k+1)})^2 \nabla \cdot \vec{u}^{n+1,(k+1)} \end{pmatrix}$$
  $E^{n+1,(k+1)} \leftarrow E^{*,(k+1)} - \Delta t \frac{\nabla \cdot (\vec{u}p)^{n+1,(k+1)}}{\rho^{n+1,(k+1)}}$
  $k \leftarrow k + 1$
end while

The advection commands in Algorithm1 are discretized using the directionally split cell integrated semi-Lagrangian advection algorithm with piecewise

constant reconstruction[26, 47] (see Figures 1, 2, and 3). The relation between the variables' control volumes and the computational mesh are shown in Figures 1, 2, and 3). The piecewise constant representation guarantees that no new extrema are generated for $\vec{u}^*$ and E^*, the updated density ρ^{n+1} is positive, and the discrete mass, momentum, and energy are conserved.

For incompressible flows, it has already been shown that the staggered grid discretization [33] of steps 9 and 10 in algorithm 1 results in a provably stable and convergent method.

For the compressible case, if we consider a simplified problem, in which the staggered grid discretization of steps 9 and 10 in algorithm 1 is applied to the wave equation:

$$\rho \vec{u}_t = -\nabla p \qquad \rho = \rho(\vec{x}) \ \rho \text{ is positive} \tag{6}$$

$$\frac{1}{\rho c^2} p_t = -\nabla \cdot \vec{u} \qquad c = c(\vec{x}) \ c \text{ is positive} \tag{7}$$

then the staggered grid approach guarantees unconditional stability for any fixed $\frac{\Delta t}{\Delta x}$.

1.2 Lax diffusion for Energy and Momentum equations in elastic materials

In this section, we just focus on single material compressible elastic fluid dynamics with a general equation of state and periodic boundary conditions. The derivations here are applicable to the general multimaterial case.

The compressible elastic fluid dynamic equations[51, 37, 11] are:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{8}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I} - s) = 0 \tag{9}$$

$$(\rho E)_t + \nabla \cdot (\rho \vec{u} E + p\vec{u} - \vec{u} \cdot s) = 0 \tag{10}$$

$$E = e + \frac{1}{2} \vec{u} \cdot \vec{u} \tag{11}$$

$$(\rho s)_t + \nabla \cdot (\rho \vec{u} s) = \rho(\Omega s - s\Omega + 2G(\bar{D} - D^p)) \tag{12}$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \tag{13}$$

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \tag{14}$$

$$p = p(\rho, e) \qquad c^2 = p_\rho + \frac{p_e p}{\rho^2} \qquad e = c_v T \tag{15}$$

D^p is the plastic strain rate and is defined below (39).

The elastic tensor s is discretized on a staggered grid as illustrated in Figures 1, 2, and 3, and as recommended by [50, 48, 42].

We discretize the elastic term, $\nabla \cdot s$ in (9), and the $\nabla \cdot (\vec{u}p)$ term in (10) using central differencing for which we have found to cause problems in elastic materials for realistic values of the shear modulus G. In order to overcome this difficulty, we add Lax Friedrichs Diffusion[32] to both the momentum equation (9),

$$\nabla \cdot (2\mu_{\text{Lax}} D), \tag{16}$$

$$\mu_{\text{Lax}} = \frac{\rho_{\max} \Delta x^2}{2\text{DIM}\Delta t} \qquad D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \qquad \text{DIM} = 2 \text{ or } 3, \tag{17}$$

and the energy equation (10),

$$\nabla \cdot (k_{\text{Lax}} \nabla T), \tag{18}$$

$$k_{\text{Lax}} = \frac{\rho_{\max} c_v \Delta x^2}{2\text{DIM}\Delta t}. \tag{19}$$

An outline of our asymptotically preserving temporal discretization of non-isothermal fluid dynamics for an elastic material, including Lax-Friedrichs smoothing, is given by Algorithm 2.

Remarks:

1. In [32], the artificial diffusion term(s) are discretized explicitly in time, but in our implementation, diffusion is discretized in time semi-implicitly; see steps 9 and 10 in Algorithm 2.

2. In [32], it is proved for general scalar hyperbolic problems that the Lax method converges if the CFL condition is met. For systems of hyperbolic conservation laws, Lax conjectures that the method converges, but does not offer a proof. We have a similar conjecture, that Lax artificial diffusion guarantees convergence; our conjecture is stronger. We conjecture that adding Lax artificial diffusion to our method guarantees convergence so long as the "velocity" CFL condition and "shear modulus" CFL conditions are met (see section 3.1), but sound speed restrictions on the time step are unnecessary.

3. We have done extensive testing of our numerical method and have found that Lax artificial diffusion to be necessary only for stiff elastic materials. We have found that adding Lax diffusion to liquid or gas materials would impose excessive damping.

Algorithm 2 Asymptotically Preserving time discretization for an elastic material

$k \leftarrow 0$

$\vec{u}^{n+1,(k)} \leftarrow \vec{u}^{n}$

while $k \leq 1$ **do**

$\rho^{n+1,(k+1)} \leftarrow \rho^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u}^{n+1,(k)}) dt$

$\rho^{n+1,(k+1)} \vec{u}^{*,(k+1)} \leftarrow (\rho \vec{u})^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho \vec{u} \otimes \vec{u}^{n+1,(k)}) dt$

$\rho^{n+1,(k+1)} E^{*,(k+1)} \leftarrow (\rho E)^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho E \vec{u}^{n+1,(k)}) dt$

$\rho^{n+1,(k+1)} s^{*,(k+1)} \leftarrow (\rho s)^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot (\rho s \vec{u}^{n+1,(k)}) dt$

$s^{n+1,(k+1)} \leftarrow s^{*,(k+1)} + \Delta t (\Omega^{*,(k+1)} s^{*,(k+1)} - s^{*,(k+1)} \Omega^{*,(k+1)} + 2G(\bar{D}^{*,(k+1)} - (D^{p})^{*,(k+1)}))$

$\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{*,(k+1)} + \Delta t \frac{\nabla \cdot (2(\mu + \mu_{\mathrm{Lax}}) D^{**,(k+1)})}{\rho^{n+1,(k+1)}}$

$T^{**,(k+1)} \leftarrow T^{*,(k+1)} + \Delta t \frac{\nabla \cdot ((k + k_{\mathrm{Lax}}) \nabla T^{**,(k+1)})}{\rho^{n+1,(k+1)} c_v}$

$\vec{u}^{***,(k+1)} \leftarrow \vec{u}^{**,(k+1)} + \Delta t \frac{\nabla \cdot s^{*,(k+1)}}{\rho^{n+1,(k+1)}}$

$T^{***,(k+1)} \leftarrow T^{**,(k+1)} + \Delta t \frac{\nabla \vec{u}^{***,(k+1)} : s^{*,(k+1)}}{\rho^{n+1,(k+1)} c_v}$

$p^{*,(k+1)} \leftarrow p(\rho^{n+1,(k+1)}, e^{***,(k+1)})$

$c^{*,(k+1)} \leftarrow c(\rho^{n+1,(k+1)}, e^{***,(k+1)})$

$$\begin{pmatrix} \vec{u}^{n+1,(k+1)} \\ p^{n+1,(k+1)} \end{pmatrix} \leftarrow \begin{pmatrix} \vec{u}^{***,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\rho^{n+1,(k+1)}} \\ p^{*,(k+1)} - \Delta t \rho^{n+1,(k+1)} (c^{*,(k+1)})^2 \nabla \cdot \vec{u}^{n+1,(k+1)} \end{pmatrix}$$

$E^{n+1,(k+1)} \leftarrow E^{***,(k+1)} - \Delta t \frac{\nabla \cdot (\vec{u} p)^{n+1,(k+1)}}{\rho^{n+1,(k+1)}}$

$k \leftarrow k + 1$

end while

4. We are only aware of proofs of convergence for the Godunov method and fractional step Lax-Friedrichs method for isentropic gas dynamics[60]. Neither of these methods are asymptotically preserving.

5. Our numerical algorithm never requires the solution of nonlinear algebraic equations, thereby avoiding robustness problems in which a discrete solution cannot be found or avoiding problems in which the discrete solution is not unique.

2 Equations for Multiphase and Multimaterial flows interacting with elastic materials.

We assume that we have M materials total consisting of M_e elastic materials and M_f fluid materials. Referring to Figure 5, the elastic materials occupy the regions in which the elastic level set function[40], ϕ_{m_e}, is positive:

$$\Omega_{m_e}(t) = \{(\vec{x}, t) | \phi_{m_e}(\vec{x}, t) \geq 0\} \tag{20}$$

A fluid material, m_f, occupies the region in which all of the elastic level set functions[40] are negative, and the fluid material level set function is positive:

$$\Omega_{m_f}(t) = \left\{(\vec{x}, t) | \phi_{m_e}(\vec{x}, t) < 0 \text{ and } \phi_{m_f}(\vec{x}, t) > 0\right\} \tag{21}$$

Referring to Figure **??**, we discretize the interface(s) using the "continuous moment of fluid method"[63]. The volume fraction and centroid in a given computational cell, Ω, are defined as:

$$F_m = \frac{\int_\Omega H(\phi_m) d\vec{x}}{|\Omega|} \tag{22}$$

$$\vec{x}_m = \frac{\int_\Omega \vec{x} H(\phi_m) d\vec{x}}{|\Omega| F_m} \tag{23}$$

$H(\phi)$ is the Heaviside function defined as:

$$H(\phi) = \begin{cases} 1 & \phi \geq 0 \\ 0 & \phi < 0 \end{cases} \tag{24}$$

2.1 Governing Equations in the bulk regions of a given material

In the bulk regions of a given material we have the following equations for compressible materials[26]:

$$\rho_t + \nabla \cdot (\rho \vec{u}) = 0 \tag{25}$$

$$(\rho \vec{u})_t + \nabla \cdot (\rho \vec{u} \otimes \vec{u} + p\mathbb{I} - \tau) = \vec{F}_{body} \tag{26}$$

$$(\rho E)_t + \nabla \cdot ((\rho E + p)\vec{u} - \vec{u} \cdot \tau - k\nabla T) = \vec{u} \cdot \vec{F}_{body} \tag{27}$$

$$\tau = 2\mu \bar{D} \tag{28}$$

$$\bar{D} = D - \frac{\text{Trace}(D)\mathbb{I}}{\text{Trace}(\mathbb{I})} \tag{29}$$

$$D = \frac{\nabla \vec{u} + (\nabla \vec{u})^T}{2} \tag{30}$$

$$\rho E = \rho e_{\text{int}} + \frac{1}{2}\rho |\vec{u}|^2 \tag{31}$$

$$p = p(\rho, e) \qquad e = c_v T \qquad c^2 = p_\rho + \frac{p_e p}{\rho^2} \tag{32}$$

If the material is a viscoelastic FENE-CR material, then one has the additional equation for the configuration tensor A[49],

$$(\rho A)_t + \nabla \cdot (\rho \vec{u} A) = \rho(A \cdot (\nabla \vec{u})^T + \nabla \vec{u} \cdot A - \frac{f(R)}{\lambda}(A - \mathbb{I})), \tag{33}$$

and the momentum constitutive law is modified,

$$\tau = \tau + \frac{\tilde{\eta}_P}{\lambda} f(R) A. \tag{34}$$

If the material is an elastic material undergoing plastic deformation, then one has the following additional equation for the deviatoric stress tensor[51, 52, 37]

$$(\rho s)_t + \nabla \cdot (\rho \vec{u} s) = \rho(\Omega s - s\Omega + 2G(\bar{D} - D^p)), \tag{35}$$

and the momentum constitutive law is modified,

$$\tau = \tau + s. \tag{36}$$

If the material is an elastic material undergoing plastic deformation, then the plastic strain equation is solved too,

$$\epsilon^p_t + \vec{u} \cdot \nabla \epsilon^p = \sqrt{\frac{2}{3}} |\Lambda|. \tag{37}$$

$$|\Lambda| = \sqrt{\frac{3}{2}} \sqrt{D^p : D^p} \tag{38}$$

$$D^p = \Lambda N \quad N = \sqrt{3/2}\frac{s}{S_e} \quad (\dot{\epsilon}^p)^2 = \frac{2}{3}\Lambda^2 \tag{39}$$

Ω is the spin tensor,

$$\Omega_{ij} = \frac{1}{2}(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}) \tag{40}$$

The effective stress is defined as,

$$S_e = \sqrt{\frac{3}{2}}\sqrt{s:s} \tag{41}$$

The Johnson-Cook model yield stress is given by,

$$\sigma_y = (A + B(\epsilon^p)^n)(1 + C\ln(\frac{\dot{\epsilon}^p}{\dot{\epsilon}_0}))(1 - \theta^m) \quad \theta = \frac{T - T_0}{T_m - T_0} \tag{42}$$

One form of the Zerilli Armstrong model yield stress is given by,

$$\beta = \beta_0 - \beta_1 \log(\dot{\epsilon}^p/\dot{\epsilon}^p_{ref}) \tag{43}$$

$$\sigma_y = A + B(\epsilon^p)^n e^{-\beta T} \tag{44}$$

The energy contribution due to elasto-plastic deformation is modeled as,

$$\beta\dot{\epsilon}^p S_e, \tag{45}$$

and is added to (27). In order to determine Λ, as it appears in (39), a "radial return algorithm" is implemented as given by Algorithm 3. We refer the reader also to [51] for the case when $n = 1$ in (42) or (44).

2.2 Governing Equations at immiscible interfaces

At an interface separating two materials, m_{source} and m_{dest}, with possible phase change[41, 55, 54, 53, 63], the interface velocity is either:

$$\vec{u}_\Gamma = \vec{u}_{m_{source}} + \frac{|\dot{m}|}{\rho_{m_{source}}}\vec{n}_\Gamma \tag{46}$$

or

$$\vec{u}_\Gamma = \vec{u}_{m_{dest}} + \frac{|\dot{m}|}{\rho_{m_{dest}}}\vec{n}_\Gamma. \tag{47}$$

$\vec{n}_\Gamma$ is the interface normal vector pointing from material m_{dest} to material m_{source}.

Algorithm 3 Radial return algorithm for determining the plastic strain rate tensor D^p.

$k \leftarrow 0$
$\dot{\epsilon}^p \leftarrow 0$
$S_e \leftarrow \sqrt{\frac{3}{2}}\sqrt{s : s}$
while $k \leq 1$ **do**
    $\sigma_y \leftarrow \sigma_y(T, \epsilon^p, \dot{\epsilon}^p)$
    if $S_e \leq \sigma_y$ **then**
        $\Lambda \leftarrow 0$, $D^p \leftarrow 0$
    else
        Solve for ϵ^p_{new}: $S_e - \sigma_y = 3G(\epsilon^p_{new} - \epsilon^p) + B((\epsilon^p_{new})^n - (\epsilon^p)^n)$
        $\dot{\epsilon}^p \leftarrow \frac{\epsilon^p_{new} - \epsilon^p}{\Delta t}$
        $\Lambda \leftarrow \sqrt{\frac{3}{2}}\dot{\epsilon}^p$
        $D^p \leftarrow \Lambda N$
    end if
    $k \leftarrow k + 1$
end while

The rate of mass transfer is,

$$\dot{m} = \frac{k_{m_{source}} \nabla T_{m_{source}} - k_{m_{dest}} \nabla T_{m_{dest}}}{L} \cdot \vec{n}_\Gamma. \tag{48}$$

The phase change interface temperature condition is,

$$T_{m_{source}} = T_{m_{dest}} = T_{\text{saturation}}(P_{\text{equilibrium}}) \tag{49}$$

Let the tensor σ represent pressure, viscosity, viscoelastic, and elastic contributions to the momentum equations. One has:

$$\sigma = -p\mathbb{I} + \tau. \tag{50}$$

The stress jump conditions are

$$\vec{n}_\Gamma \cdot (-\sigma_{m_{sourse}} + \sigma_{m_{dest}}) \cdot \vec{n}_\Gamma = -\gamma\kappa_\Gamma - \dot{m}^2(\frac{1}{\rho_{m_{source}}} - \frac{1}{\rho_{m_{dest}}}) \tag{51}$$

$$(\vec{t}_{i,tangent})_{m_{sourse}} \cdot (-\sigma_{m_{source}} + \sigma_{m_{dest}}) \cdot \vec{n}_\Gamma = 0 \quad i = 1, 2 \tag{52}$$

The interface curvature is,

$$\kappa_\Gamma = \nabla \cdot \vec{n}_\Gamma \tag{53}$$

3 Discretization for multiphase flows interacting with elastic materials

In Algorithm 4 we give the overall temporal discretization of our asymptotically preserving multiphase flow method. The Algorithm 4 is very similar to that described in [26] except for the following key differences:

1. The present algorithm is staggered. Please refer to Figures 1, 2, and 3. A staggered algorithm has better stability properties than a collocated method[33, 50, 48, 23].

2. In order to accurately allow for droplets and/or bubbles to interact with elastic materials, we have made the following modifications to [26]:

   (a) In steps 5,6 and 7, the elastic state variables are advected using an "extended" velocity field. Please refer to Figure 6.

   (b) Because the elastic force (step 22, Algorithm 4) is applied with a smoothed Heaviside function, we must make sure that the deviatoric stress tensor s_{m_e} is well defined in a narrow band just outside of material m_e: $\phi_{m_e} > -\alpha\Delta x$. This is achieved by steps 14 and 15 of Algorithm 4. Please see Figure 6. In our numerical experiments, we always have $\alpha = 4$.

   (c) As recommended by [48], the viscous force and pressure gradient force are first approximated without the inclusion of surface tension; please see Steps 21 through 25 of Algorithm 4. Then the viscous force and pressure gradient force are recomputed in steps 27 through 29 in which the surface tension force is included and the velocity in the elastic material regions is fixed to the velocity after step 25.

   (d) the static or dynamic contact angle is enforced by way of extending the level set function and volume of fluid functions from the fluid regions into the elastic regions; please see step 8 of Algorithm 4. The extension algorithm is the same as that described in our earlier work[6] except that in the present article, we are considering flexible substrates whereas in [6] the substrate only underwent rigid motion.

Remarks:

1. Steps 25 and 29 of Algorithm 4 corresponds to solving the following, linear, pressure equation,

$$-\frac{p-\tilde{p}}{\rho c^2 \Delta t^2} + \nabla \cdot \frac{\nabla p}{\rho} = \frac{\nabla \cdot \vec{u}^*}{\Delta t}, \tag{54}$$

   where $p \equiv p^{n+1,(k+1)}$ and $\vec{u}^{**,(k+1)} \equiv \vec{u}^*$.

2. The phase change algorithm[54, 53, 63], given by steps 11 and 12 in Algorithm 4, is described in more detail in Algorithm 5. Please also see Figure 4.

3. The multimaterial reinitialization algorithm, given by step 10 in Algorithm 4, is described in detail in [63].

4. The flotsam removal algorithm, given by step 9 in Algorithm 4, is described in more detail in Algorithm **??**.

5. The algorithm for deriving the face centered density and face centered pressure (satisfying constant contact condition[31]) needed in steps 22, 25, 26, and 29, of Algorithm 4, is described in more detail in Algorithm **??**. Please also see Figure 5.

6. The algorithm for deriving the face centered surface tension force, needed in step 29 of Algorithm 4, is described in more detail in our earlier work[35].

7. A property of our numerical method is that there is no longer a "contact line singularity"[65, 1] at triple junctions of gas, liquid, and elastic materials. Since the elastic material is deformable (very slightly if the physical properties warrant!) at triple junctions, the mathematical model, and numerical algorithm allows for both the no-slip condition and contact line dynamics to exist simultaneously.

3.1 Time step stability restrictions

The time step, Δt, must satisfy the following conditions[63, 35].

1. **CFL condition** In order that characteristics do not collide, the following CFL condition holds:

$$\Delta t \leq \frac{1}{2} \frac{\min \Delta x}{\max |\vec{u}|_\infty}$$

2. **Phase change CFL condition**

$$\Delta t \leq \frac{1}{2} \frac{\min \Delta x}{\max |\vec{u}^{\text{phase change}}|_\infty}$$

3. **Surface tension stability constraint**

$$\Delta t \leq \frac{1}{2} (\min \Delta x^{3/2}) \sqrt{\frac{\rho_\Gamma}{2\pi\sigma_\Gamma}}$$

ρ_Γ is given by Algorithm 8 when $(\nabla H)_\Gamma \neq 0$.

4. **Elastic material stability constraint**

$$U_{elastic} = \sqrt{\frac{8s : s}{\rho G}}$$

$$\Delta t \leq \frac{1}{2} \frac{\min \Delta x}{\max |U_{elastic}|}$$

Remark:

In contrast to [63], we choose the EI-LE advection algorithm[26, 47] rather than the Weymouth and Yue algorithm[58], since the EI-LE has a less stringent time step constraint. The drawback of using EI-LE is that it is not exactly volume preserving in 3D; but we have found the errors to be a fraction of a percent (see section **??**).

Algorithm 4 Multiphase flows interacting with elastic materials

$k \leftarrow 0$

$\vec{u}^{n+1,(k)} \leftarrow \vec{u}^{n}$

while $k \leq 1$ **do**

$\vec{U} \leftarrow (\rho, \rho\vec{u}, \rho E, \rho s)^T \quad \vec{\mathcal{F}} \leftarrow (F, \phi)$

$\vec{V}_m \leftarrow \begin{cases} \vec{u}^{n+1,(k)} & \text{if } m \text{ is fluid material} \\ \text{Extend}(\vec{u}^{n+1,(k)}) & \text{if } m \text{ is elastic material} \end{cases}$

$\vec{\mathcal{F}}_m^{n+1,(k+1)} \leftarrow \vec{\mathcal{F}}_m^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot (\vec{\mathcal{F}}_m \otimes \vec{V}_m) - (\nabla \cdot \vec{V}_m)\vec{\mathcal{F}}_m dt \quad m \in [1, M]$

$(F\vec{U})_m^{*,(k+1)} \leftarrow (F\vec{U})_m^{n} + \int_{t^n}^{t^{n+1}} \nabla \cdot ((F\vec{U})_m \otimes \vec{V}_m) dt \quad m \in [1, M]$

$\phi_m^{n+1,(k+1)} \leftarrow \text{Extend}(\phi_m^{n+1,(k+1)})$ ▷ from fluid region into elastic

$F_m^{n+1,(k+1)} \leftarrow \text{Remove Flotsam}(F_m^{n+1,(k+1)})$

Reinitialize $\phi_m^{n+1,(k+1)}$ to be signed distance.

$\phi_{m_{source}} \leftarrow \phi_{m_{source}} - \Delta t \frac{|\dot{m}|}{\rho}$

$\phi_{m_{dest}} \leftarrow \phi_{m_{dest}} + \Delta t \frac{|\dot{m}|}{\rho}$

for $m = 1, \ldots, M_e$ **do**

$\vec{u}^{extend} \leftarrow \text{Extend}(\vec{u}^{*,(k+1)})$ ▷ from elastic region m to outside

$s^{extend} \leftarrow \text{Extend}(s_m^{*,(k+1)})$ ▷ from elastic region m to outside

$\Omega_{ij} \leftarrow \frac{1}{2}(u_{i,j}^{extend} - u_{j,i}^{extend})$

$D_{ij} \leftarrow \frac{1}{2}(u_{i,j}^{extend} + u_{j,i}^{extend})$

$s_m^{n+1,(k+1)} \leftarrow s^{extend} + \Delta t(\Omega s^{extend} - s^{extend}\Omega + 2G_m(\bar{D} - D^p))$

end for

$T^{**,(k+1)} \leftarrow T^{*,(k+1)} + \Delta t \frac{\nabla \cdot ((k + k_{\text{Lax}}) \nabla T^{**,(k+1)})}{\rho^{n+1,(k+1)} c_v}$

$\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{*,(k+1)} + \Delta t \frac{\nabla \cdot (2(\mu + \mu_{\text{Lax}}) D^{**,(k+1)})}{\rho^{n+1,(k+1)}}$

$\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{**,(k+1)} + \Delta t \frac{\sum_{m=1}^{M_e} \nabla \cdot (H_{\alpha\Delta x}(\phi_m) s_m^{*,(k+1)})}{\rho^{n+1,(k+1)}}$

$T_m^{**,(k+1)} \leftarrow T_m^{**,(k+1)} + \Delta t \frac{\nabla \vec{u}^{**,(k+1)} : \tau_m}{\rho_m^{n+1,(k+1)} c_{v,m}} \quad m \in [1, M]$

$\tilde{\rho} \leftarrow \rho^{n+1,(k+1)} \quad \tilde{p} \leftarrow p(\tilde{\rho}, e^{**,(k+1)}) \quad c^2 \leftarrow c^2(\tilde{\rho}, e^{**,(k+1)})$

$$\begin{pmatrix} \vec{u}^{n+1,(k+1)} \\ p^{n+1,(k+1)} \end{pmatrix} \leftarrow \begin{pmatrix} \vec{u}^{**,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\tilde{\rho}} \\ \tilde{p} - \Delta t \tilde{\rho} c^2 \nabla \cdot \vec{u}^{n+1,(k+1)} \end{pmatrix}$$

$E^{n+1,(k+1)} \leftarrow E^{**,(k+1)} - \Delta t \frac{\nabla \cdot (\vec{u}p)^{n+1,(k+1)}}{\tilde{\rho}}$

$\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{*,(k+1)} + \Delta t \frac{\nabla \cdot (2(\mu + \mu_{\text{Lax}}) D^{**,(k+1)})}{\tilde{\rho}}$ ▷ $\vec{u}_{m_e}$ clamped

$\vec{u}^{**,(k+1)} \leftarrow \vec{u}^{**,(k+1)} - \Delta t \frac{\sum_{\Gamma} \gamma_{\Gamma} \kappa_{\Gamma} \nabla H_{\Gamma}}{\tilde{\rho}}$

$$\begin{pmatrix} \vec{u}^{n+1,(k+1)} \\ p^{n+1,(k+1)} \end{pmatrix} \leftarrow \begin{pmatrix} \vec{u}^{**,(k+1)} - \Delta t \frac{\nabla p^{n+1,(k+1)}}{\tilde{\rho}} \\ \tilde{p} - \Delta t \tilde{\rho} c^2 \nabla \cdot \vec{u}^{n+1,(k+1)} \end{pmatrix}$$ ▷ $\vec{u}_{m_e}$ clamped

$k \leftarrow k + 1$

end while

Algorithm 5 Phase change algorithm

1: For each cell in a narrow band about the interface, use the closest point map to find the corresponding closest point on the interface. Approximate the phase change speed, $|\dot{m}|/\rho$ at the closest point. (Figure 4)
2: extend the phase change speed $|\dot{m}|/\rho$ further into bulk regions.
3: Advect the volume fractions using unsplit CISL advection with velocity $|\dot{m}|\vec{n}_\Gamma/\rho$.
4: The overall mass correction in order to conserve mass due to expansion or compression is: $M_{cor} = \sum(F_{dest}^{new} - F_{dest}^{old})V_{grid}(\rho_{m_s} - \rho_{m_d})$.
5: If the interface velocity is in the form $\vec{u}_I = \vec{u}_{m_d} + \frac{|\dot{m}|}{\rho_{m_d}}\vec{n}_\Gamma$, then distribute M_{cor} at least two cells deep into the source material m_s.
6: If the interface velocity is in the form $\vec{u}_I = \vec{u}_{m_s} + \frac{|\dot{m}|}{\rho_{m_s}}\vec{n}_\Gamma$, then distribute M_{cor} at least two cells deep into the destination material m_d.
7: If the material with the distributed mass, M_{cor}, is an incompressible material, then M_{cor} becomes a source term for the velocity projection equation, otherwise, the density is incremented accordingly.

Algorithm 6 Algorithm for determining the unique material number associated with a given grid cell

Require: Level set functions ϕ_m for $m = 1, \ldots, M$.
Ensure: $m_{primary}$
1: $m_e \leftarrow \max\phi_m$ $\quad$ m is elastic
2: **if** $\phi_{m_e} \geq 0$ **then**
3: $\quad m_{primary} \leftarrow m_e$
4: **else**
5: $\quad m_{\max} \leftarrow \max\phi_m$ $\quad$ m is not elastic
6: $\quad m_{primary} \leftarrow m_{\max}$
7: **end if**

Algorithm 7 Algorithm for determining $(\nabla H_\Gamma)_{i+1/2,j}$ at cell face $(i+1/2, j)$

Require: Level set functions $(\phi_m)_{i,j}$ for $m = 1, \ldots, M$.
Require: Level set functions $(\phi_m)_{i+1,j}$ for $m = 1, \ldots, M$.
Ensure: $(\nabla H_\Gamma)_{i+1/2,j}$
m_L obtained from $(\phi_m)_{i,j}$ ▷ Algorithm 6
m_R obtained from $(\phi_m)_{i+1,j}$ ▷ Algorithm 6
if m_L is an elastic material **then**
    $(\nabla H_\Gamma)_{i+1/2,j} \leftarrow 0$
else if m_R is an elastic material **then**
    $(\nabla H_\Gamma)_{i+1/2,j} \leftarrow 0$
else if $m_R == m_L$ **then**
    $(\nabla H_\Gamma)_{i+1/2,j} \leftarrow 0$
else if $m_R < m_L$ **then**
    $(\nabla H_\Gamma)_{i+1/2,j} \leftarrow \frac{1}{\Delta x}$
else
    $(\nabla H_\Gamma)_{i+1/2,j} \leftarrow -\frac{1}{\Delta x}$
end if

Algorithm 8 Algorithm for determining the face centered density, $\rho_{i+1/2,j}$ at cell face $(i+1/2, j)$

Require: Level set functions $(\phi_m)_{i,j}$ for $m = 1, \ldots, M$. ▷ Figure 5
Require: Level set functions $(\phi_m)_{i+1,j}$ for $m = 1, \ldots, M$. ▷ Figure 5
Require: Tessellating interface reconstruction in cell (i, j) ▷ Figure 5
Require: Tessellating interface reconstruction in cell $(i+1, j)$ ▷ Figure 5
Require: refined density $(\rho_m)_{2i+1,2j}, \ldots, (\rho_m)_{2i+2,2j+1}$ for $m = 1, \ldots, M$ ▷ Figure 5
Ensure: $\rho_{i+1/2,j}$
$\text{mass}^L_{i+1/2,j} \leftarrow \sum_{m=1}^{M} \sum_{j_f=2j}^{2j+1} |\Omega_{ij} \cap \Omega_{i+1/2,j} \cap \Omega_{m,ij}| (\rho_m)_{2i+1,j_f}$
$\text{mass}^R_{i+1/2,j} \leftarrow \sum_{m=1}^{M} \sum_{j_f=2j}^{2j+1} |\Omega_{i+1,j} \cap \Omega_{i+1/2,j} \cap \Omega_{m,i+1,j}| (\rho_m)_{2i+2,j_f}$
$\rho_{i+1/2,j} \leftarrow (\text{mass}^L_{i+1/2,j} + \text{mass}^R_{i+1/2,j}) / |\Omega_{i+1/2,j}|$
Determine $(\nabla H_{m_L,m_R})_{i+1/2,j}$, m_L, and m_R from Algorithm 7.
if $\gamma_{m_L,m_R} \nabla H_{m_L,m_R} \neq 0$ **then**
    $\rho_{i+1/2,j} \leftarrow \max(\rho_{i+1/2,j}, \theta\rho_{m_L}, \theta\rho_{m_R})$
end if

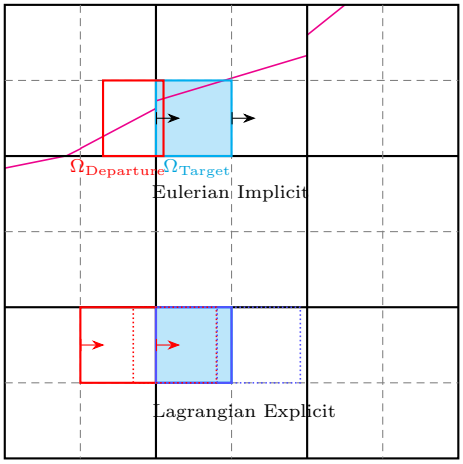


Figure 1: Illustration of the Eulerian Implicit, Lagrangian Explicit cell integrated semi-Lagrangian (CISL) method[26, 47]. Rather than have separate traversals through the grid for the cell-centered, face centered, and edge centered variables, we instead interpolate all variables to a refined grid, followed by one traversal through the refined grid, and then conservatively average the respective results back to their defined control volumes.

4 Results

4.1 Drop impact on flexible substrate, sensitivity to contact angle condition and substrate shear modulus

Kim et al[29]

Water droplet.

Weber number equals twenty

$$\text{We} == \frac{\rho U_0^2 D_o}{\gamma} \tag{55}$$

Superhydrophobic advancing contact angle is $\theta_A > 150^0$.

PDMS flexible plate material. Density of flexible plate is $1.0 \frac{\text{g}}{\text{cm}^3}$.

Young's modulus $E = 2.5E + 6$ Pascals.

Plate thickness is one millimeter. Plate horizontal dimensions are 40 by 32 millimeters; clamped on the 40 millimeter side.

Simulation plate thickness is 1 millimeter and simulation (cylindrical) plate radius is 10 millimeters.

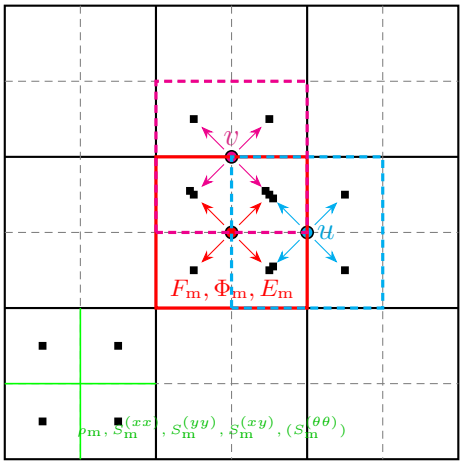


Figure 2: The material volume fractions, F_m, level set functions, ϕ_m, and total energy, E_m, are stored at cell centers and then consistently interpolated to the refined grid in preparation for Cell Integrated Semi-Lagrangian advection[26, 47]. The velocity (u, v) is piecewise constant interpolated from the face centered (MAC grid) control volumes to the refined grid cell centers. The horizontal advective velocity, u^{advect}, is linearly interpolated to the refined grid face centers in the horizontal direction and is piecewise constant interpolated in the vertical direction. The vertical advective velocity, v^{advect}, is linearly interpolated in the vertical direction and piecewise constant interpolated in the horizontal direction. The material density ρ_m and deviatoric stress tensor components, $s_m^{xx,xy,yy,\Theta\Theta}$ (RZ), are already discretized on the refined grid.

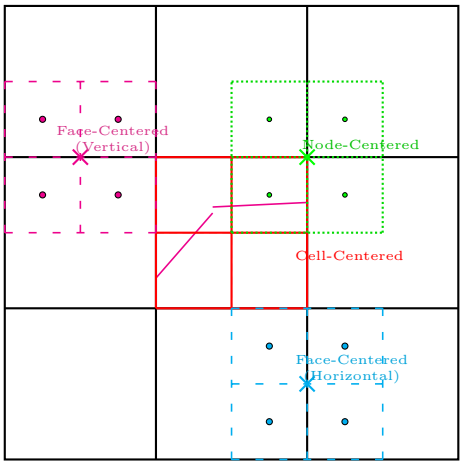


Figure 3: After Cell Integrated Semi-Lagrangian advection[26, 47] on the refined grid, the cell centered variables, F_m, ϕ_m, and E_m, and the face centered variables, (u, v), are conservatively "recombined" back to their respective control voume locations. The deviatoric stress tensor components, $s_m^{xx,yy,\Theta\Theta}$ (RZ), are conservatively recombined to the cell centered control volumes, and then piecewise constant prolongated back to the refined grid. The off-diagonal deviatoric stress tensor component, s_{xy}, is conservatively recombined to the node control volume (RZ) (edge centers in 3D), and then piecewise constant prolongated back to the refined grid. The material density, ρ_m, do not need to be recombined.

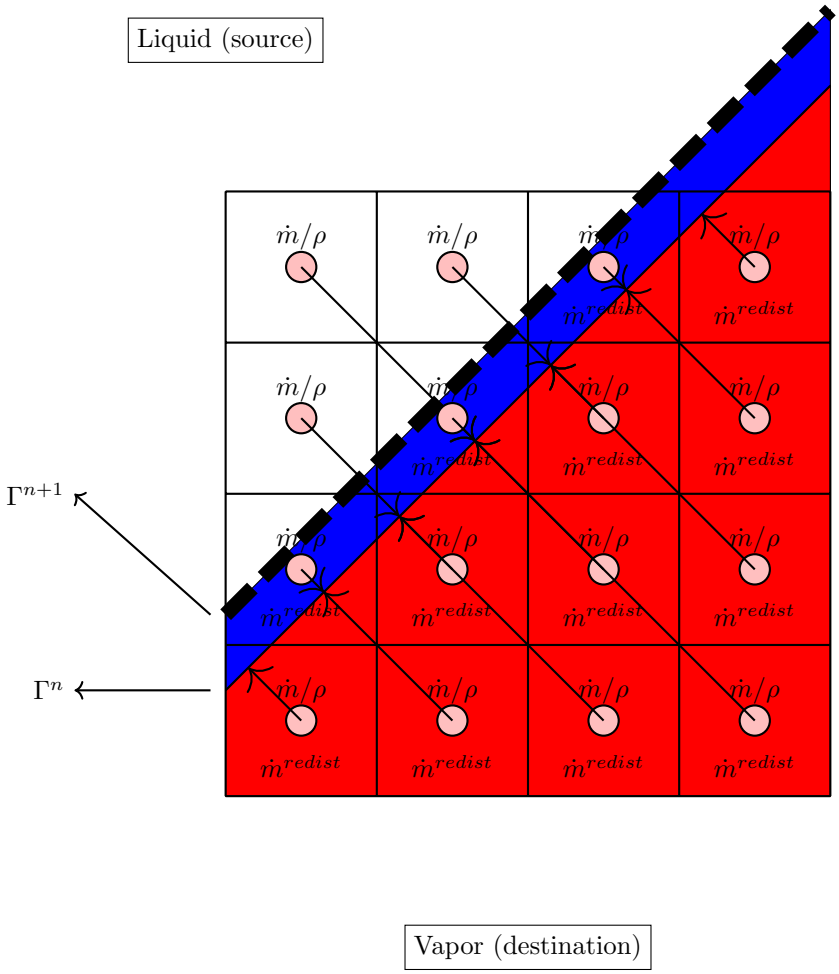


Figure 4: The phase change algorithm: (1) the phase change speed, $|\dot{m}|/\rho$, is approximated at all narrow band cells by evaluating $|\dot{m}|/\rho$ at the corresponding closest point on the interface (Γ^n). (2) The change in volume fraction due to unsplit CISL advection determines the overall mass correction, M_{cor}, due to expansion or compression, $M_{cor} = \sum(F_{dest}^{new} - F_{dest}^{old})V_{grid}(\rho_{m_s} - \rho_{m_d})$. M_{cor} becomes a source term for the projection step if the material is incompressible, or directly increments the density if the material is compressible. (3) M_{cor} is distributed to the vapor (destination) side of the Γ^{n+1} interface if the interface speed is $\vec{u}_{m_s} + \frac{|\dot{m}|}{\rho_{m_s}}\vec{n}_\Gamma$, and M_{cor} is distributed to the liquid (source) side of the interface if the interface speed is $\vec{u}_{m_d} + \frac{|\dot{m}|}{\rho_{m_d}}\vec{n}_\Gamma$. Referring to the diagram, if distributing M_{cor} to the vapor side, M_{cor} must be all distributed beyond the blue region into the red region.

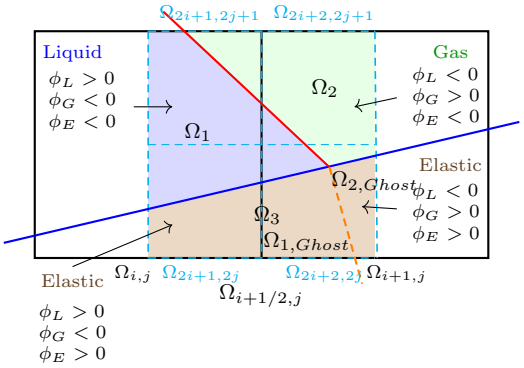


Figure 5: Example for determining the mixture density in face centered control volumes. In this case, the face is $(i+1/2, j)$. Momentum conservation is guaranteed since the density for each material is discretized on the refined grid $(2i+1, 2j)$, $(2i+1, 2j+1)$, $(2i+2, 2j)$, and $(2i+2, 2j+1)$. The Fluid regions tessellate the computational domain (Ω_1, $\Omega_{1,Ghost}$, Ω_2, $\Omega_{2,Ghost}$) and the elastic material regions (Ω_3) are embedded within the computational domain. Prior to computing mixture density, we use the nested dissection algorithm in order to convert the material reconstruction so that Fluid and Elastic materials tessellate the computational domain (Ω_1, Ω_2, Ω_3).

4.2 Drop impact on rigid substrate and freezing with ice modeled as elastic material. Sensitivity to the Weber number

Jin et al[28]

Water droplet.

Weber number equals 123.0 Difference in substrate temperature and freezing temperature is $\Delta T = 64^0$ Kelvin.

According to [16], the viscosity of water at 223^0 Kelvin is 17.0 centipoise (one poise equals 0.1 Pascal $\cdot$ second). Surface tension at 240^0 Kelvin is 80 dynes per square centimeter. Static angle of 65^0 imposed between liquid and ice or liquid and substrate.

$$\text{We} == \frac{\rho U_0^2 D_o}{\gamma} \tag{56}$$

4.3 Impact of a Tungsten rod with a steel plate.

[51]

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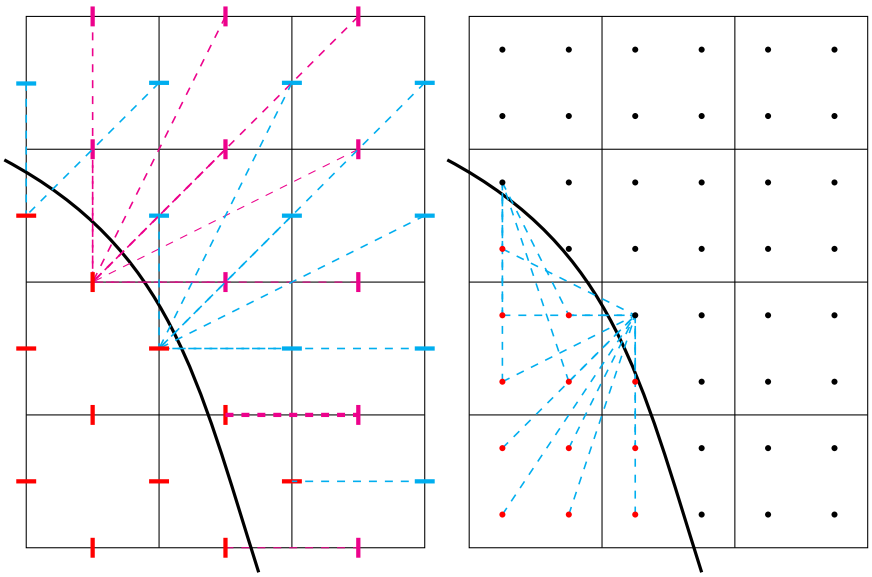


Figure 6: Left: The elastic velocity field is derived from the overall "fluid" velocity field by extrapolating the fluid velocity from within elastic materials (at least one adjoining cell to a face element has a positive elastic level set value) to the outside of elastic materials. The closest elastic material velocity is determined from 9^3 stencils about "non-elastic" velocities. In this illustration, magenta corresponds to vertical fluid velocities, cyan corresponds to horizontal fluid velocities, and red is the elastic material velocity. Right: The inverse distance squared weighted elastic material configuration tensor value is determined from 9^3 stencils about the surrounding "non-elastic" sub-cells. In this illustration, red points correspond to the elastic material.

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